



Unit → 3

→ Transformer → AC device

A device which transfers power (AC) from one circuit to another. Working of the transformer is based on Faraday's law of electromagnetic induction.

- It's a static device
- It transfers power without changing frequency

Transformer

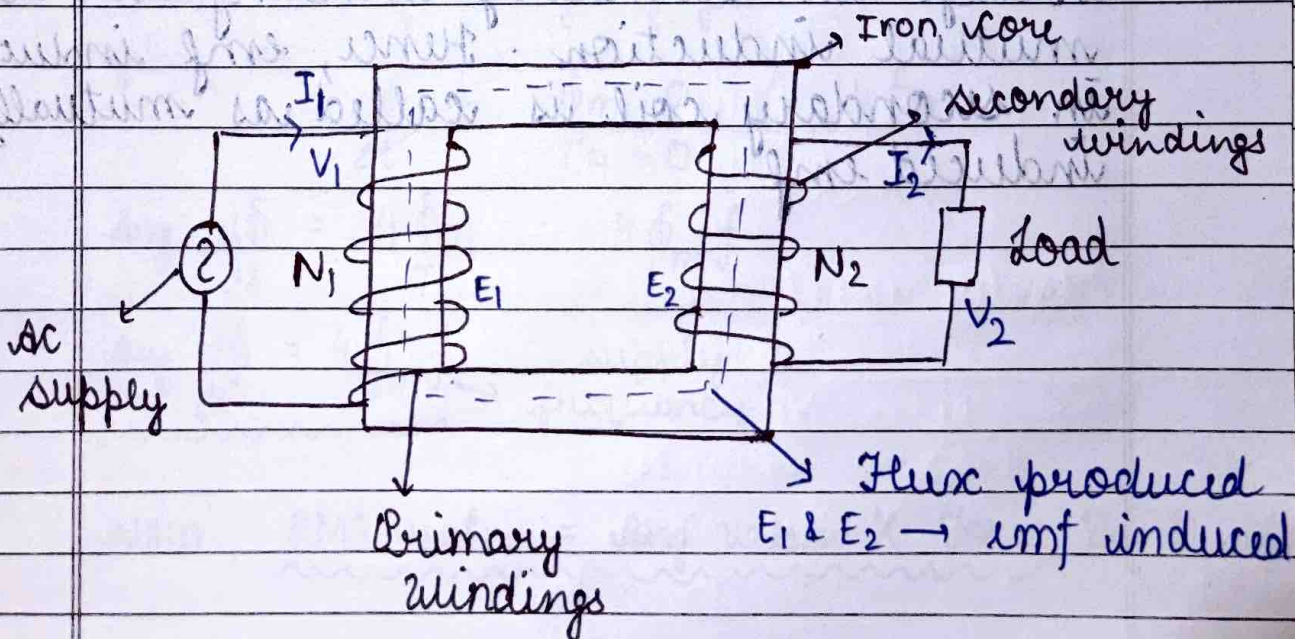
Step Up

$$V_{out} > V_{in} \\ N_2 > N_1$$

Step Down

$$V_{out} < V_{in} \\ N_2 < N_1$$

- Transformers are designed for either single phase or three phase supply. However principle of operation remains same.
 Single phase → domestic purpose
 Three phase → industrial purpose





→ Structure of Transformer

- It consists of 2 highly inductive coils (windings) wound on iron/steel core.
- Coil connected to AC supply is primary winding whereas other one is secondary winding.
- Both windings are isolated from each other as well as from iron core.

→ Working of transformer

As soon as primary winding is connected to single phase AC supply, an AC current starts flowing through it. This primary AC current produces an alternating flux (ϕ) in primary side of core.

Most of this changing flux gets linked with secondary side of core. This varying flux will induce voltage in secondary winding acc to Faraday's law of electromagnetic induction. Thus due to primary current there is induced voltage in secondary winding due to mutual induction. Hence, emf induced in secondary coil is called as mutually induced emf.



→ EMF equation

Let V_1 = Primary voltage

V_2 = Secondary voltage

E_1 = Primary induced EMF

E_2 = Secondary induced EMF

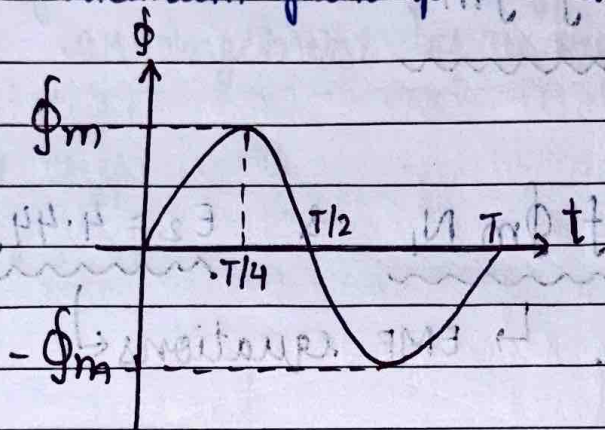
N_1 = Primary windings

N_2 = Secondary windings

I_1 = Primary current

I_2 = Secondary current

ϕ_m = Maximum flux produced in magnetic core



Now, by Faraday's Law

$$E = N \frac{d\phi}{dt}$$

So for both E_1 and E_2

$$E_1 = N_1 \frac{d\phi}{dt} \quad \text{and} \quad E_2 = N_2 \frac{d\phi}{dt}$$

From sine wave, flux changes from 0 to ϕ_m in time 0 to $T/4$ sec

∴ Avg rate of change of flux w.r.t time can be given as

$$= \frac{\Delta\phi}{\Delta t} = \frac{\phi_m - 0}{T/4 - 0} = \frac{4\phi_m}{T}$$

$$\text{Avg } \frac{d\phi}{dt} = \frac{4\phi_m}{T} = 4\phi_m f$$

$$\Rightarrow \text{Avg } \frac{d\phi}{dt} = 4\phi_m f \rightarrow \text{supply frequency}$$

Now, RMS value = Avg value X Form Factor

$$\therefore \text{RMS } \frac{d\phi}{dt} = \text{Avg } \frac{d\phi}{dt} \times 1.11$$

$$\Rightarrow \text{RMS } \frac{d\phi}{dt} = \frac{d\phi}{dt} = 4 \times 1.11 \times \phi_m f$$

$$\Rightarrow \frac{d\phi}{dt} = 4.44 f \phi_m$$

Hence,

$$E_1 = 4.44 f \phi_m N_1 \quad \& \quad E_2 = 4.44 f \phi_m N_2$$

↳ EMF equations

Also,

$$\phi_m = B_m \times A$$

$$\Rightarrow E_1 = 4.44 f B_m A N_1 \quad \& \quad E_2 = 4.44 f B_m A N_2$$

→ Transformation Ratio (Voltage current relat)

Since,

$$E_1 = 4.44 f B_m A N_1 \quad \text{--- (1)}$$

$$\text{and } E_2 = 4.44 f B_m A N_2 \quad \text{--- (2)}$$

÷ (1) & (2)

$$\Rightarrow \frac{E_1}{E_2} = \frac{N_1}{N_2}$$

Under ideal cond, $E_1 = -V_1$ & $E_2 = -V_2$

$$\Rightarrow \frac{E_1}{E_2} = \frac{V_1}{V_2} = \frac{N_1}{N_2}$$

$$\text{Also } I_1 N_1 = I_2 N_2$$

$$\Rightarrow \frac{I_2}{I_1} = \frac{N_1}{N_2}$$

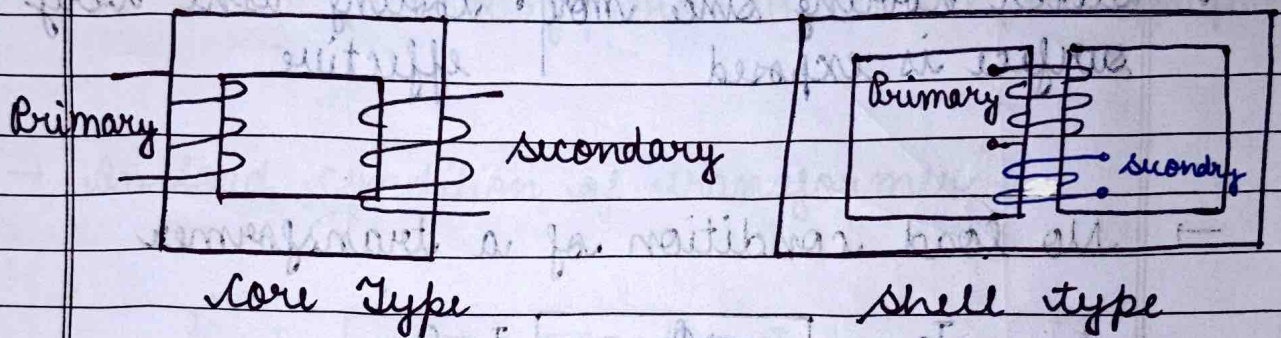


$$\frac{E_1}{E_2} = \frac{V_1}{V_2} = \frac{N_1}{N_2} = \frac{I_2}{I_1} = K \quad \text{Transformation ratio}$$

→ Types of transformers

Transformers are of different types depending upon arrangement of core & windings.

1. Core type
2. Shell type
3. Berry type



→ Core Type Transformer

- The core of this transformer is in the form of rectangular frame made from laminations.
- It provides single magnetic circuit.
- Primary & secondary windings are uniformly distributed on two limbs of core.
- Both windings are of cylindrical shape and are arranged in a concentric manner with low voltage windings placed near the core.

→ Shell type transformer

- Primary & secondary windings are placed on the central limb of core. The high & low voltage windings are of sandwich type in form of

interleaved pancakes.

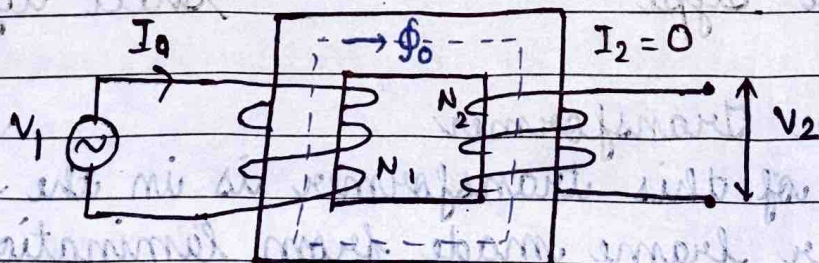
This provides two magnetic circuits

- It provides better mechanical support and protection for the windings

→ Core Type vs Shell type

- | Core Type | Shell type |
|--|--------------------------------|
| • Core has only 1 window | • Core has 2 windows |
| • Windings encircle core | • Core encircles winding |
| • Cylindrical windings | • Sandwiched windings |
| • Easy to repair | • Not easily repaired |
| • Less mechanical protection | • Better mech. protection |
| • Better cooling since more surface is exposed | • Cooling isn't very effective |

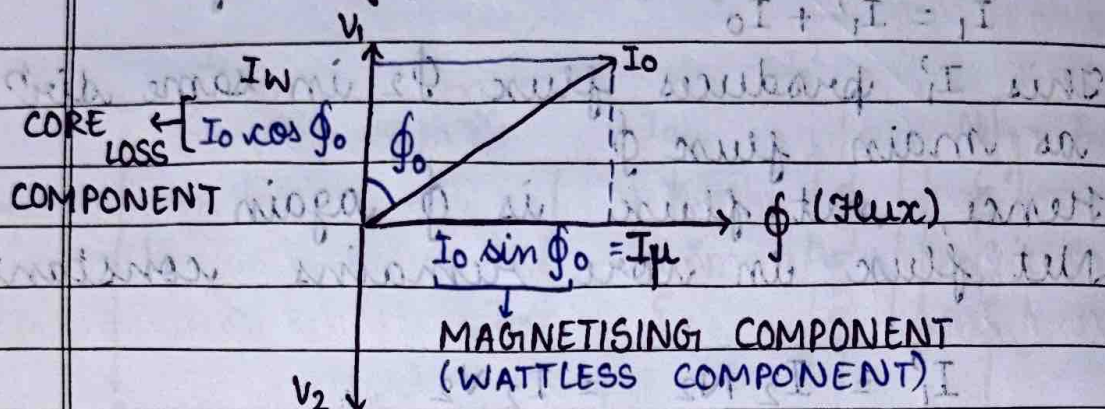
→ No load condition of a transformer



When no load is connected across secondary winding then secondary current $I_2 = 0$. Hence, I_2 must be 0, but in practical transformer, a very small current I_0 (NO. LOAD CURRENT) flows in primary side. This current I_0 has to supply iron loss in the core and very small copper loss and produces the flux in (Φ_0) core i.e. to magnetise the core.

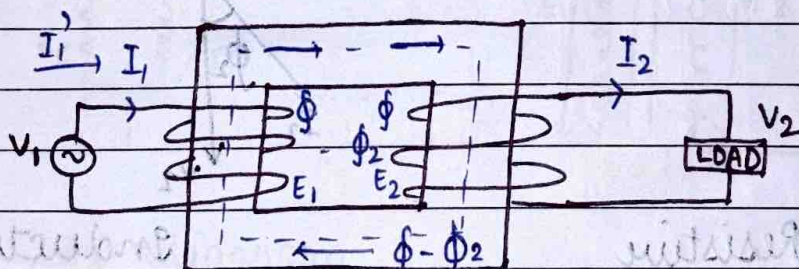


→ Phasor diagram in no load condition



- I_m maintains flux in core
- I_w supplies iron loss & small copper loss and is active / working / iron loss / core loss component

→ On load condition of transformer



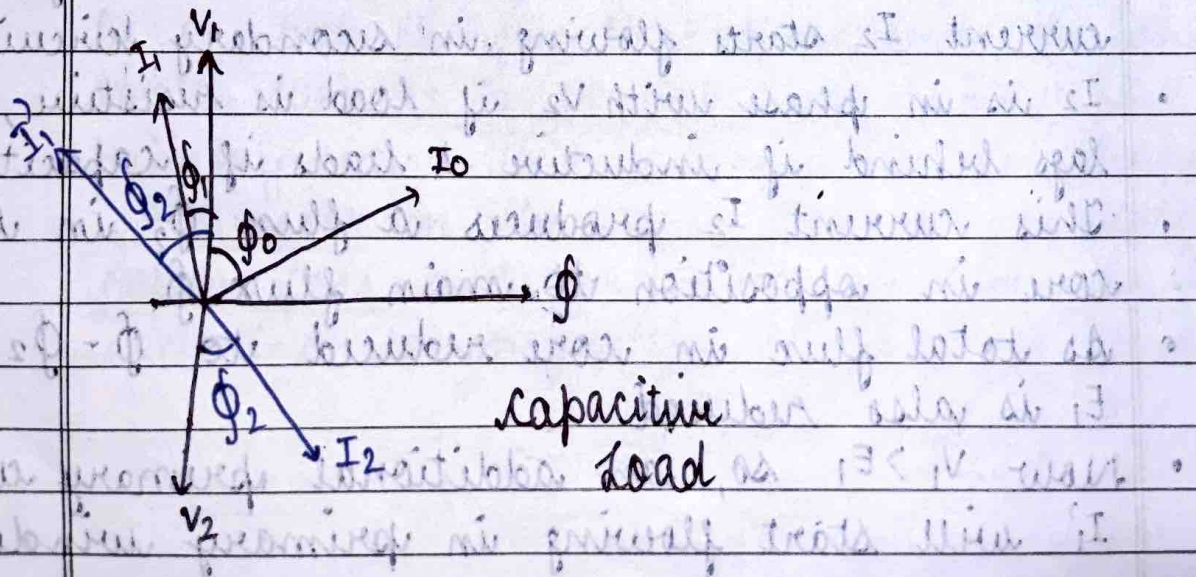
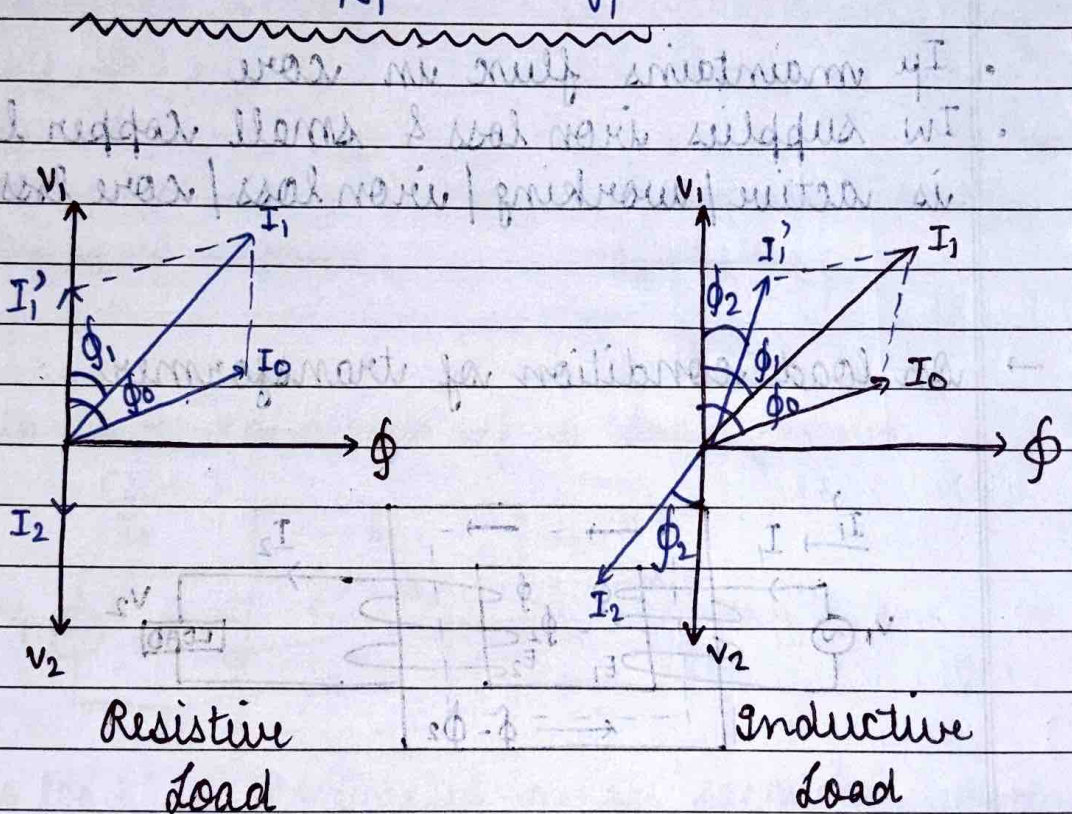
- When the load is connected across secondary terminals, current I_2 starts flowing in secondary circuit.
- I_2 is in phase with V_2 if load is resistive, lags behind if inductive & leads if capacitive.
- This current I_2 produces a flux Φ_2 in the core in opposition to main flux Φ .
- As total flux in core reduced to $\Phi - \Phi_2$ so E_1 is also reduced.
- Now $V_1 > E_1$ so, an additional primary current I_1' will start flowing in primary winding.



- Now, total current in primary is

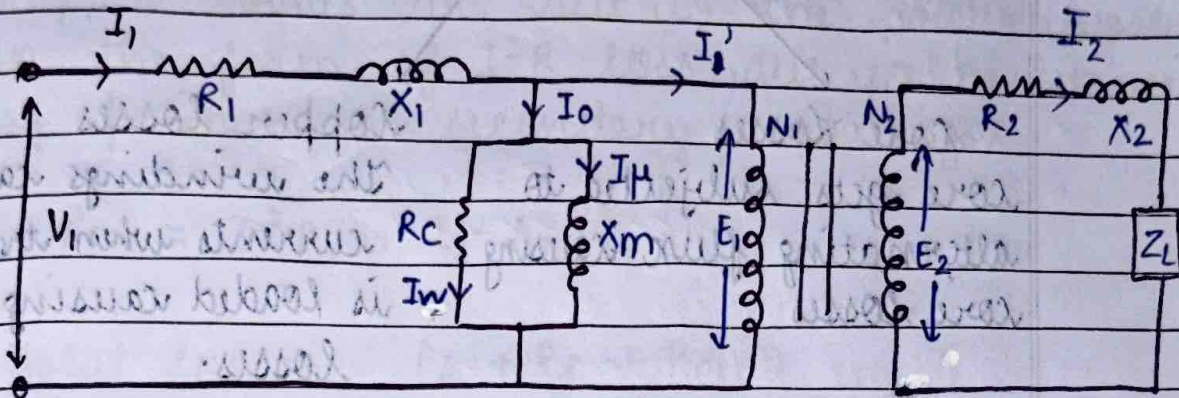
$$I_1 = I_1' + I_0$$
- This I_1' produces flux ϕ_2 in same dirⁿ as main flux ϕ
- Hence, net flux is ϕ again
- Net flux in core remains constant

$$I_1' \frac{N_1}{N_2} = I_2 \frac{N_2}{N_1} \Rightarrow I_1' = I_2 \frac{N_2^2}{N_1^2}$$





→ Equivalent circuit diagram

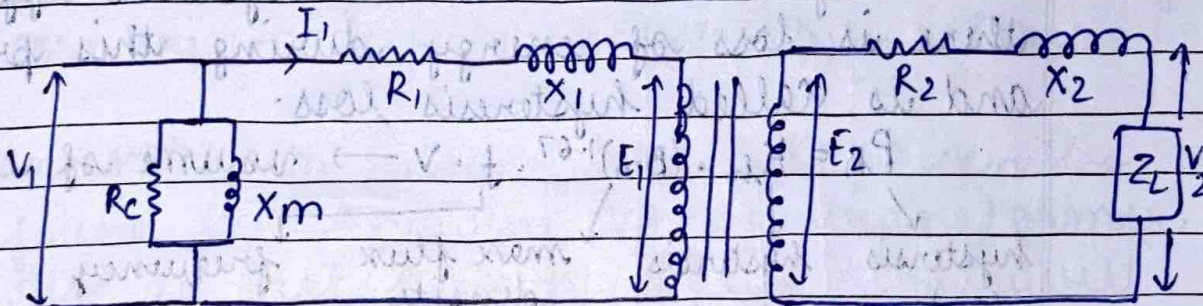


$$\vec{I}_1 = \vec{I}_0 + \vec{I}_1'$$



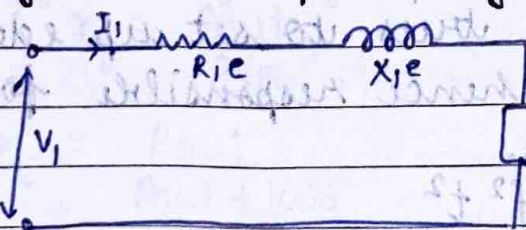
No load primary current (very low)

$$\vec{I}_1 \approx \vec{I}_1'$$



→ eq ckt diagram

• Referred to primary

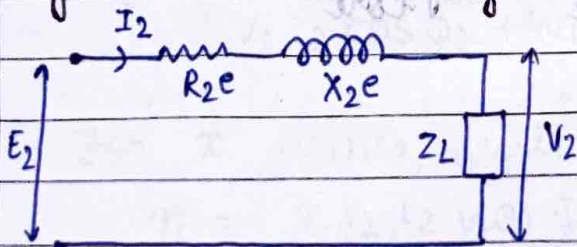


$R_{1e} \rightarrow$ eq resis refer to prim

$$R_{1e} = R_1 + R_2 \left(\frac{N_1}{N_2} \right)^2$$

$$X_{1e} = X_1 + X_2 \left(\frac{N_1}{N_2} \right)^2$$

• Referred to secondary



$$R_{2e} = R_2 + R_1 \left(\frac{N_2}{N_1} \right)^2$$

$$X_{2e} = X_2 + X_1 \left(\frac{N_2}{N_1} \right)^2$$

→ Losses in a transformer

Core losses

Core gets subjected to alternating flux causing core losses

Copper losses

The windings carry currents when transformer is loaded causing copper losses.

→ Core / Iron losses (constant losses)

• Hysteresis loss

Due to AC flux set up in magnetic core of transformer, it undergoes cycle of magnetisation or demagnetisation. Due to hysteresis effect there is loss of energy during this process and is called hysteresis loss.

$$P_h = K_h \cdot (B_m)^{1.67} \cdot f \cdot V \rightarrow \text{volume of core}$$

$\swarrow$ hysteresis loss $\downarrow$ hysteresis constant $\swarrow$ max flux density $\swarrow$ frequency

• Eddy current losses

Induced emf in core tries to set up eddy current in core & hence responsible for the eddy current losses.

$$P_e = K_e \cdot B_m^2 \cdot f^2 \cdot t^2$$

$\swarrow$ eddy current loss $\downarrow$ eddy current constant $\downarrow$ thickness of core



- copper losses / variable losses
copper losses are due to the power wasted in the form of I^2R loss due to resistance of primary & secondary windings.

$$P_{cu} = I_1^2 R_1 + I_2^2 R_2$$

$$\text{Total losses} = P_h + P_e + P_{cu}$$

Q. Why rating of transformer is in units of VA or KVA?

Rating of any device depends on losses occur in that device.

- Cu loss → depends on current via windings
- Core loss → depends on supply voltage

In order to know total losses can be found by knowing VA of transformer. Also it's not known which load will be used so VA ratings are favoured.

→ efficiency of transformer

$$\eta = \frac{P_{out}}{P_{in}}$$

$$\eta = \frac{P_{out}}{P_{out} + \text{loss}}$$

• FOR FULL LOAD

$$\eta = \frac{V_2 I_2 \cos \phi_2}{V_2 I_2 \cos \phi_2 + W_i + W_{cu}}$$

For x times full load

$$\eta = \frac{x V_2 I_2 \cos \phi}{x V_2 I_2 \cos \phi + P_i + x^2 P_c}$$



→ working

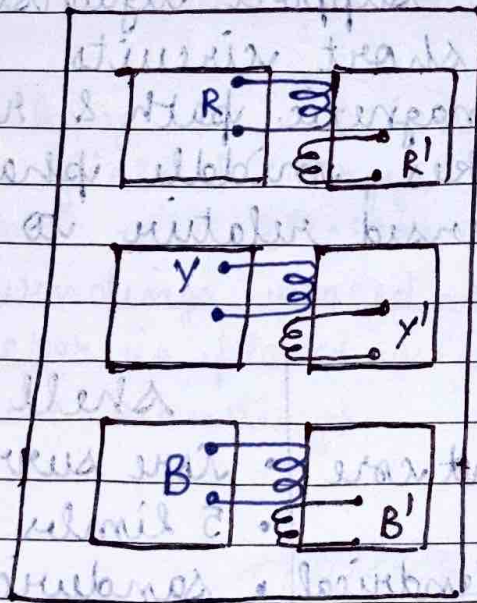
- The 3 limbs are physically joined by top & bottom yokes

- when 3- ϕ power is applied, fluxes ϕ_R , ϕ_Y & ϕ_B are produced in 3 limbs

at any instant, the sum of fluxes in a balanced 3 ϕ system is 0. $\phi_R + \phi_Y + \phi_B = 0$

Therefore, any 2 limbs acts as return path for flux of third limb. loop

→ shell type 3 ϕ transformer



construction

A shell type transformer can be thought as of 3 individual single phase transformers placed together.

The core surrounds windings like shells. It typically has 5 limbs (3 main limbs for the windings & 2 outer limbs to provide additional return paths for flux).

It uses sandwich windings where LV & HV



coils are stacked in layers rather than being concentric cylinders.

- working bridge is made of 8 mm.
- The flux produced by each phase winding is divided. Half of flux from central limb travels through yoke to adjacent limb and other half travels to outer part of shell.
- Because the magnetic circuits are relatively independent for each phase, shell type provides better support against mechanical stress during short circuits.
- To optimize magnetic path & reduce the height of yokes, middle phase winding is often reversed relative to outer phases.

→ Core Type

- windings around core
- 3 limbs
- concentric cylindrical windings
- easily repairable
- Better natural cooling
- Low resistance to magnetic forces
- Used for high voltage power applications

Shell Type

- Core surround winding
- 5 limbs
- sandwiched or interleaved windings
- difficult to repair
- Less effective cooling
- High mechanical strength
- used for low voltage high current applications



DC Machines

→ generator

→ motor

generator

I/P

Mechanical energy

O/P

Electrical energy

motor

Electrical energy

Mechanical energy

Machine

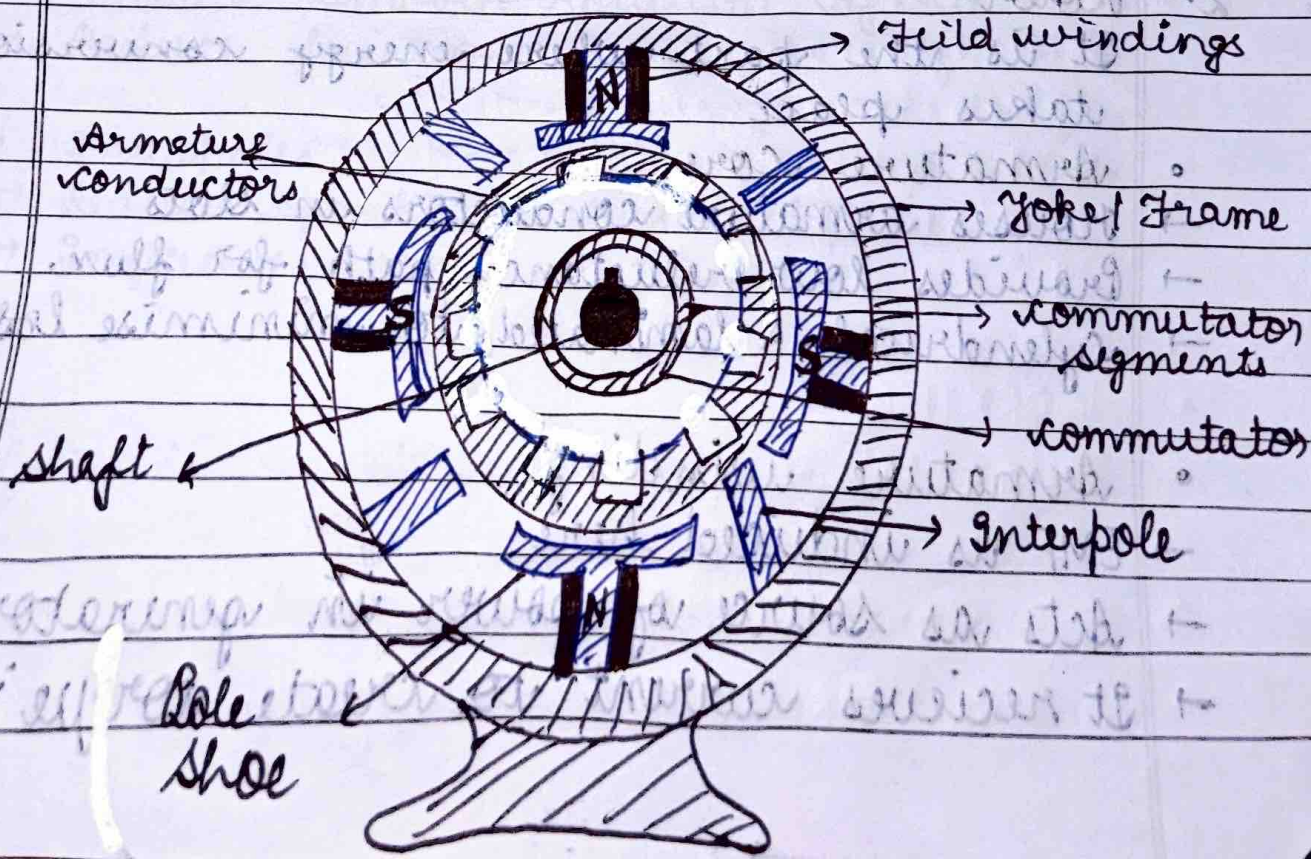
Stator Part

Rotor Part

Remains in static position

Free to rotate

For making DC machines, we require conductor, magnetic field & energy. To produce a magnetic field, we use electromagnet which have the field windings placed on stator and the conductor is placed on rotor to make the armature windings.





→ Parts of the DC machine

1. Stator

It provides magnetic housing & mechanical support for machine

• Yoke

→ It serves as outer protective covering

→ It supports magnetic poles

→ Provides low reluctance path for flux

• Pole shoe

→ Carry field windings to produce magnetic flux

→ Spreads flux uniformly in air gaps

→ Support field windings

• Field windings

→ Become electromagnetically charged when current is passed

→ Create main magnetic field for machine

2. Rotor

It is the part where energy conversion takes place

• Armature core

→ Houses armature conductors in slots

→ Provides low reluctance path for flux

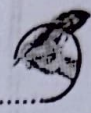
→ Cylindrical & laminated to minimise losses

• Armature windings

→ EMF is induced here

→ Acts as source of power in generator

→ It receives current to create torque in motor



• Shaft

- Transfers mechanical energy to or from the machine

3. Commutators

It's responsible for managing direction of current

• commutator

- Acts as mechanical rectifier
- In generator, converts AC induced in armature into DC for external circuit
- In motor, reverses current direction in coils as they pass under different poles to maintain unidirectional torque.

• Brushes

- maintain physical contact with rotating commutator to collect / supply current

• Bearings

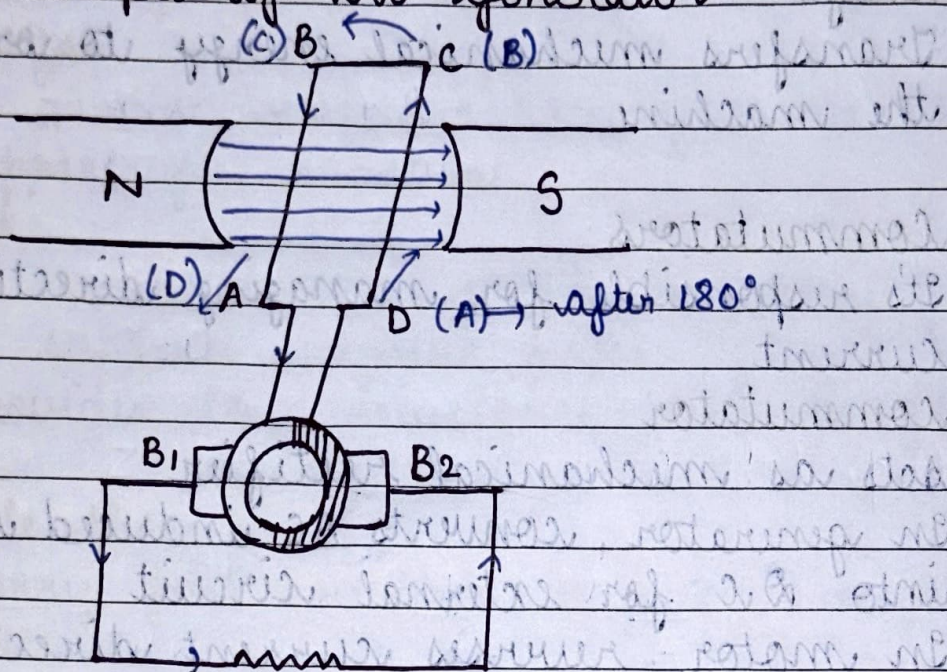
- Ensure smooth rotation of rotor
- Reduce friction

• Interpoles

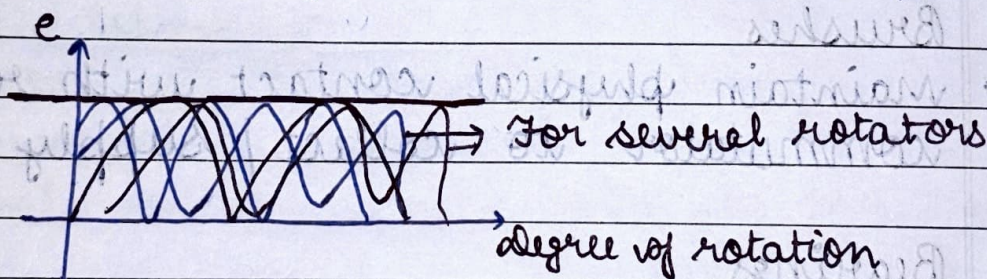
- Reduce sparking at brushes
- Improves commutation



→ Principle of AC generator



After 180° posⁿ of ABCD changes but direction of current remains same



- **PRINCIPLE.** AC generator operates on Faraday's law of electromagnetic induction. It states that whenever a conductor cuts the magnetic flux, an EMF is induced in it.

• WORKING

1. Field coils are energised to create magnetic field
2. Armature is rotated by external prime mover like turbine / engine

3. As armature conductors rotate, they cut magnetic field lines.
4. Acc to Fleming's Right hand rule, an EMF is induced in conductors.
5. While EMF is naturally alternating, the commutator & brushes ensures current flows in 1 dirⁿ in external circuit.

→ EMF EQUATIONS

Let

P = no of poles of generator

ϕ = Flux produced by each pole

N = Speed of armature rpm

Z = Total no of armature conductors

A = No of parallel paths in which total no of conductors are divided

For lap type winding, $A = P$

For wave type winding, $A = 2$

By Faraday's law, $e = \frac{d\phi}{dt}$ (single conduc)

Consider 1 revolution of a conductor, it will cut total flux produced by poles

$$= P\phi$$

Flux cut by conductor in 1 revolution

$$d\phi = P\phi \text{ Weber}$$

Time req to complete 1 revolution

$$dt = \frac{60}{N} \text{ sec}$$



Hence avg value of emf induced in single conductor

$$e = \frac{d\phi}{dt} = \frac{P\phi}{60N}$$

$$= \frac{P\phi N}{60}$$

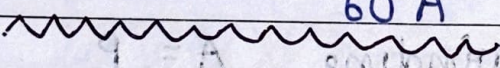
For Z conductors in parallel paths

$\Rightarrow$ Total emf

$$E = \frac{P\phi N}{60} \times \frac{Z}{A}$$

$\therefore$ all are in series & emf remains same across all paths

$$\Rightarrow \text{EMF} = \frac{P\phi N Z}{60 A}$$



$\rightarrow$ D.C Motor

Electrical $\rightarrow$ Mechanical

When conductor placed in B, it exp. a force (Lorentz force law)

Flux due to armature interacts with flux of electricity magnetic field

$O \rightarrow G \rightarrow$ emf induced, oppose ext $V \rightarrow$ Back emf